

## MACHINE LEARNING FUNDAMENTALS SERIES

# Lasso (L1) Regression

*Feature Selection, L1 Regularization & Sparse Modeling*

## 1 Introduction and Problem Motivation

In machine learning workflows, fitting linear models via Ordinary Least Squares (OLS) minimizes training error by finding parameters that minimize the Sum of Squared Residuals (SSR) [2]. However, when datasets have small sample sizes ( $n$ ) or contain many non-informative/useless predictor variables ( $p$ ), standard OLS models tend to severely **overfit** the training data [2, 5].

Overfitting manifests as **High Variance**: while OLS achieves near-zero training error, its predictions degrade significantly when evaluated on unseen testing data [2, 5].

**Lasso Regression** (Least Absolute Shrinkage and Selection Operator) resolves this problem by introducing an **L1 Regularization Penalty** [2, 6]. By trading a small, controlled amount of *bias*, Lasso produces a massive reduction in model *variance*, improving long-term prediction accuracy and performing automatic **Feature Selection** [2, 5].

## 2 Mathematical Formulation and Objective Function

Lasso Regression replaces the quadratic penalty of Ridge Regression with an absolute value penalty [2, 6].

### 2.1 The Lasso Loss Function

For a continuous response variable  $y_i$  and  $p$  predictor variables  $x_{i1}, x_{i2}, \dots, x_{ip}$  across  $n$  observations, Lasso Regression minimizes the following penalized loss function [2, 3]:

$$\mathcal{L}_{\text{Lasso}}(\beta) = \sum_{i=1}^n \left( y_i - \left( \beta_0 + \sum_{j=1}^p \beta_j x_{ij} \right) \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \quad (1)$$

where:

- $\text{SSR} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$  is the **Sum of Squared Residuals** measuring goodness-of-fit [2].
- $\lambda \geq 0$  is the **Regularization Hyperparameter** (tuning parameter) controlling penalty intensity [2].
- $\sum_{j=1}^p |\beta_j| = \|\beta_{1:p}\|_1$  is the **L1 Norm** penalty on feature slope coefficients [2, 6].

In matrix notation, the objective is expressed as [2, 6]:

$$\min_{\beta} \left\{ \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta_{1:p}\|_1 \right\} \quad (2)$$

**Technical Note: Exclusion of the Y-Intercept ( $\beta_0$ )**

Just like in Ridge Regression, the L1 penalty term  $\sum_{j=1}^p |\beta_j|$  starts at  $j = 1$  and **explicitly excludes the y-intercept**  $\beta_0$  [3]. The intercept  $\beta_0$  represents the baseline expected value of  $y$  when all features  $x_j = 0$ . Penalizing  $\beta_0$  would artificially force model predictions towards zero regardless of the actual mean scale of the target variable  $y$  [3].

### 3 Why Lasso Can Shrink Slopes Exactly to Zero

The most fundamental difference between Ridge (L2) and Lasso (L1) regression is that **Lasso can shrink slope parameters  $\beta_j$  all the way to 0**, whereas Ridge shrinks them asymptotically close to zero but never hits exact zero [4, 6].

#### 3.1 Analytical View: Derivative of the Penalty

The distinction originates from the mathematical nature of the absolute value function versus the squared function near the origin [2, 6]:

- **Ridge Penalty ( $\lambda\beta_j^2$ ):** The derivative with respect to  $\beta_j$  is  $2\lambda\beta_j$ . As  $\beta_j \rightarrow 0$ , the gradient approaches 0. The penalty push gets progressively weaker, causing coefficients to slow down and approach zero asymptotically without ever reaching it [4].
- **Lasso Penalty ( $\lambda|\beta_j|$ ):** The derivative with respect to  $\beta_j$  is a constant  $+1$  (for  $\beta_j > 0$ ) or  $-1$  (for  $\beta_j < 0$ ). Because the gradient remains constant regardless of how small  $|\beta_j|$  becomes, Lasso exerts a relentless, uniform inward force that pushes small coefficients all the way to exact zero [4, 6].

#### 3.2 Geometric View: Constraint Regions and Corners

An intuitive geometric explanation comes from viewing regularization as a constrained optimization problem [4, 6]:

$$\min_{\beta} \text{SSR} \quad \text{subject to} \quad \|\beta\|_1 \leq t \quad (\text{Lasso}) \quad \text{vs.} \quad \|\beta\|_2^2 \leq t \quad (\text{Ridge}) \quad (3)$$

As shown in Figure 1:

- **L2 Constraint (Circle):** Has a smooth boundary. Expanding OLS error contours touch the circle at a point off the axes, where both  $\beta_1 \neq 0$  and  $\beta_2 \neq 0$  [4].
- **L1 Constraint (Diamond):** Has sharp corners located directly on the coordinate axes (where one or more  $\beta_j = 0$ ). Expanding OLS contours are far more likely to intersect a sharp corner first, locking the coefficient to **exact zero** [4, 6].

#### 3.3 Automatic Feature Selection and Sparsity

By setting coefficients of non-informative features to zero, Lasso produces **Sparse Models** [4, 5]:

$$\text{Size} = \beta_0 + \beta_1(\text{Weight}) + \beta_2(\text{Diet}) + \cancel{\beta_3(\text{Astrological Sign})} + \cancel{\beta_4(\text{Swallow Airspeed})} \quad (4)$$

Lasso automatically strips out useless terms (e.g., astrological sign or swallow airspeed in predicting size), leaving a clean, interpretable equation containing only true predictive variables [4, 5].

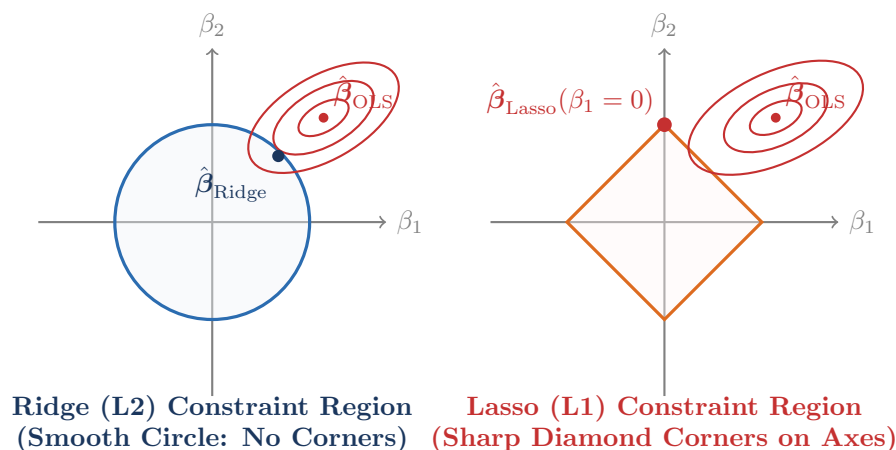


Figure 1: Geometric Comparison of Ridge (L2) and Lasso (L1) Regularization Regions. The contours represent expanding unconstrained OLS error ellipses (SSR). For Lasso, the expanding ellipse first hits a sharp corner of the diamond on the coordinate axis, forcing  $\beta_1 = 0$  exactly [4, 6].

## 4 Tuning Hyperparameter $\lambda$ via Cross-Validation

The strength of regularization is dictated by the hyperparameter  $\lambda$  [2, 4].

### 4.1 Behavior across Lambda Values

- $\lambda = 0$ : The L1 penalty vanishes ( $\lambda \sum |\beta_j| = 0$ ). Lasso becomes identical to standard OLS Least Squares (low bias, high variance) [2, 4].
- $\lambda \rightarrow \infty$ : The penalty dominates the loss function. To minimize loss, all feature slopes  $\beta_1, \dots, \beta_p$  are forced to 0. The model simplifies to a horizontal line predicting the overall sample mean  $\hat{y} = \beta_0$  [4].
- **Intermediate  $\lambda$** : As  $\lambda$  increases from 0, coefficients drop towards zero; useless features hit 0 first and are eliminated [3, 5].

### 4.2 Optimal Lambda Selection Workflow

The optimal  $\lambda^*$  value cannot be learned directly from training data; it is selected via **10-Fold Cross-Validation** [2, 4]:

1. **Data Splitting**: Divide the dataset into 10 equal-sized folds [2].
2. **Grid Search**: Define a sequence of candidate  $\lambda$  values (e.g.,  $\lambda \in [0.001, 100]$ ) [2, 4].
3. **Iterative Validation**: For each candidate  $\lambda$ :
  - Train the Lasso model on 9 folds [2].
  - Calculate validation error  $\text{SSR}_{\text{val}}$  on the remaining 1 held-out fold [2].
  - Repeat 10 times so every fold serves as the validation set once [2].
  - Compute the Average Cross-Validation Mean Squared Error ( $\overline{\text{MSE}}_{\text{CV}}$ ) [2].
4. **Optimal Point**: Select  $\lambda^*$  that achieves the global minimum  $\overline{\text{MSE}}_{\text{CV}}$ , identifying the exact balance point between bias and variance [2, 4].

## 5 Head-to-Head Comparison: Ridge (L2) vs. Lasso (L1) Regression

Both methods prevent overfitting through regularization, but they serve different modeling scenarios [2, 5, 6].

Table 1: Comprehensive Comparison between Ridge (L2) and Lasso (L1) Regularization.

| Feature / Dimension           | Ridge Regression (L2)                                    | Lasso Regression (L1)                               |
|-------------------------------|----------------------------------------------------------|-----------------------------------------------------|
| <b>Penalty Term</b>           | $\lambda \sum_{j=1}^p \beta_j^2$ (Squared L2 Norm)       | $\lambda \sum_{j=1}^p  \beta_j $ (Absolute L1 Norm) |
| <b>Derivative near 0</b>      | $2\lambda\beta_j \rightarrow 0$ (Penalty weakens near 0) | $\pm\lambda$ (Constant penalty force)               |
| <b>Slope Shrinkage</b>        | Asymptotically close to 0 ( $\beta_j \neq 0$ )           | Shrinks all the way to <b>exact 0</b>               |
| <b>Feature Selection</b>      | No (Retains all $p$ features in equation)                | <b>Yes</b> (Generates Sparse Models)                |
| <b>Constraint Region</b>      | Smooth Hypersphere / Circle                              | Diamond / Octahedron with sharp corners             |
| <b>Ideal Scenario</b>         | Most features are useful & predictive                    | <b>Many useless / noisy variables</b> present       |
| <b>Model Interpretability</b> | Lower (All variables remain in equation)                 | <b>High</b> (Excludes irrelevant variables)         |

### Core Take-away: When to Choose Which?

- **Choose Lasso Regression** when working with high-dimensional datasets containing many noisy or irrelevant features. Lasso will eliminate useless variables, simplifying the model and reducing variance [5, 6].
- **Choose Ridge Regression** when almost all features are known to have genuine predictive power. Retaining small weights for all features generally yields lower prediction variance than hard zeroing [5, 6].